RIG-SMOOTHNESS IS STABLE UNDER ADIC BASE CHANGE IN WADM^NOETH

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof chooses a common ideal of definition and delegates the rig-smoothness of the completed tensor product to the homomorphism-level base-change lemma

Editorial note. The label lemma-base-change-rig-smooth-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-smooth continuous families and adic morphisms in the category WAdm^Noeth (lemma-rig-smooth-continuous and Formal Spaces definition of adic homomorphism)
- base change of rig-smooth homomorphisms of adic rings (lemma-base-change-rig-smooth-homomorphism of the same chapter)

Source reference. Original definition 0GBR: sources/stacks-chow/tags.txt, source lines 20381–20381. Prerequisite pointer invoked at proof line(s) 4385; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0GBG: sources/stacks-chow/tags.txt, source lines 20371–20371. Prerequisite pointer invoked at proof line(s) 4376; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAM: sources/stacks-restricted/pinned-original-source.tex, source lines 1043–1043. Prerequisite pointer invoked at proof line(s) 4390; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCI: sources/stacks-restricted/pinned-original-source.tex, source lines 4214–4214. Prerequisite pointer invoked at proof line(s) 4380; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider the properties $P(\varphi)$ ="φ is rig-smooth" and $Q(\varphi)$="φ is adic" on arrows of WAdm^{Noeth}. Then P is stable under base change by Q as defined in Formal Spaces, Remark 0GBG.*

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Proof. The statement makes sense by Lemma 0GCI. To see that it is true assume we have morphisms $B \to A$ and $B \to C$ in $WAdm^{Noeth}$ and that $B \to A$ is rig-smooth and $B \to C$ is adic (Formal Spaces, Definition 0GBR). Then we can choose an ideal of definition $I \subset B$ such that the topology on A and C is the I-adic topology. In this situation it follows immediately that $A\widehat{\otimes}_B C$ is rig-smooth over (C, IC) by Lemma 0GAM. $\square$